

Supplementary material

VCF-RDFizer: From a VCF File to Explicit, Verifiable, and Policy-aware Semantic Genomic Data

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This supplement provides the implementation, inputs, protocols, and complete measurements supporting the main text. It separates completed comparisons, expected refusals, incomplete checks, and unresolved limits. The linked workflow evaluates prerequisites for the motivating applications rather than complete clinical or governance systems.

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S1 Documented capabilities of related tools

Table S1 summarizes the documented capabilities discussed in the main text and gives the evidence behind each mark. Each mark records what a tool’s public documentation states: its repository and manual, the cited paper, and package registries, inspected in September 2026 and re-checked against those sources in October 2026. A mark is not a measurement on a shared input, and an open mark means that a capability was not documented, not that it is impossible. Where a tool was superseded, the version that reads VCF was assessed. The columns are defined as follows; a partial mark applies the qualification given after each definition.

RDF model The output uses a documented vocabulary. Partial: terms shown only by example, or a vocabulary that is no longer published.

Header + genotype VCF header metadata is kept in the RDF together with per-sample genotype and FORMAT data. Partial: only one of the two, or both only through user-written templates.

Editable mappings Users can change how VCF maps to RDF without changing the converter. Partial: configuration or plug-in hooks over a fixed mapping, or mappings that produce only part of the output.

CLI Documented command-line use suitable for pipelines. Partial: library, browser, or user-interface use only.

Reproducible workflow A documented VCF-to-RDF route that a third party can rerun. Partial: manual installation of several components, an unpublished package, an offline service, or an unmaintained runtime.

Compressed RDF The tool writes compressed RDF itself. Partial: only through a bundled wrapper.

Public release A versioned package, executable, or container. Partial: source code only.

Table S1. Documented capabilities of VCF-to-RDF converters and related workflows, with the evidence for each mark. ● supported and documented; ◐ partial or limited; ○ not found in public documentation. Sources are the cited papers and the repositories, manuals, and registries named in each note.

Tool or workflow	Representation			Operation			
	RDF model	Header + genotype	Editable mappings	CLI	Reprod. workflow	Compr. RDF	Public release
<i>Dedicated VCF-to-RDF converters</i>							
VCF-RDFizer (this study)	●	●	◐	●	●	●	●
Writes the public VCF Core vocabulary, keeping the header, records, alleles, and samples (expanded or condensed). Mappings are partial: RML maps the record backbone, nine triples per record and 5.3% of the HG005 graph, while samples, INFO, QUAL, and the header come from emitters configured by options rather than mappings (Section S3.2). Pinned Docker image; HDT and COTTAS output; released on PyPI, Docker Hub, and conda-forge.							
TogoVar VCF2RDF [1]	◐	○	◐	●	●	○	●
Rust command-line tool (https://github.com/togovar/vcf2rdf) with generate , convert , and compress . Its documentation calls the RDF model very simple, shown through a configuration template. TogoVar converts allele frequencies only, for privacy, so no sample or header data. A YAML configuration selects the base IRI, namespaces, and INFO keys over fixed mapping rules. compress writes BGZF-compressed VCF, not RDF. Docker image and release binaries.							
JVarkit VcfToRdf [2]	◐	◐	○	●	●	○	●
<code>java -jar jvarkit.jar vcf2rdf</code> (documentation at https://jvarkit.readthedocs.io). A project-specific vcf: namespace is shown only by example. Output includes chromosomes, filters, samples, and genotypes (GT, DP, GQ, PL), but not INFO or FORMAT header descriptions, and the documented default of <code>--hide</code> includes GT. The only options are <code>--hide</code> and <code>-o</code> . Released as a jar and on Bioconda.							

Table S1, continued

Tool or workflow	RDF model	Header + genotype	Editable mappings	CLI	Reprod. workflow	Compr. RDF	Public release
SPARQLing Genomics [3, 4] vcf2rdf, documented in the v0.99.11 manual. Uses FALDO, Dublin Core, PROV, and its own terms, with field predicates generated from the VCF header; the manual’s namespaces are inconsistent. Every header-described field becomes triples, with per-sample calls carrying INFO and FORMAT values (only calls with an ALT allele). The mapping is fixed in code. Compression only through the <code>--compress</code> option of the bundled <code>folder2rdf</code> wrapper. Release tarball, Docker image, and Guix environment; the GitHub repository was archived in 2021 when development moved to GitLab.	●	●	○	●	●	●	●
VCF2RDF (web) [5] JavaScript for browsers and Node.js (https://github.com/ib1/VCFr); no command line is documented. Terms are minted under vcf2rdf.org/app/ , a host that is now offline, with no published schema. The paper describes header classes but not per-sample genotypes. Source code only; the web service is offline.	●	●	○	●	●	○	●
BioHackathon vcf2rdf [6, 7] Built at BioHackathon 2013 (https://github.com/fstrozzi/bioruby-vcf2rdf), with a <code>vcf2rdf</code> executable. FALDO positions with an ad hoc allele namespace. Per-sample genotypes and allele coverage are written; header metadata is not mentioned. Turtle output. The Ruby gem the README installs is not published, so only the source is available.	●	●	○	●	●	○	●
<i>General genomic-format tools</i>							
BioInterchange [8, 9] Assessed as BioInterchange 2 (v2.0.5, https://github.com/BioInterchange/BioInterchangeC); version 1, conceived at BioHackathon 2012, does not read VCF. Converts VCF 4.2 to JSON-LD in the Genomic Feature and Variation Ontology (GFVO) [10], with one metadata object per file and per-sample genotypes on each feature. Output can be changed through Python plug-in hooks and user annotations, not mapping files. RDF serializations come from JSON-LD tools; compression was applied outside the tool. Homebrew package and installer.	●	●	●	●	●	○	●
bio-vcf [11] <code>bio-vcf --rdf</code> and ERB templates such as <code>vcf2rdf.erb</code> (https://github.com/vcflib/bio-vcf) define the output, with ad hoc prefixes rather than a documented vocabulary. Header metadata and sample data are both available to templates but are combined only by the user. Released as a Ruby gem and on Debian and Bioconda.	○	●	●	●	●	○	●
ruby-rdf/rdf-vcf [12] <code>vcf2rdf file.vcf.gz</code> (https://github.com/ruby-rdf/rdf-vcf), requiring JRuby 9 and a tabix index; unmaintained since its 0.1.2 gem in September 2015. The documentation names no vocabulary and does not mention header, INFO, FORMAT, or genotype handling. Uncompressed N-Triples.	○	○	○	●	●	○	●
<i>Knowledge-graph construction pipelines</i>							
VariantKG [13] Annotates VCF with SnpEff, converts it with SPARQLing Genomics’ <code>vcf2rdf</code> (and so inherits its header and genotype coverage), then extends that vocabulary with its own ontology, including CADD scores, and links patient metadata through named graphs; no ontology file is published. Operated through a Gradio interface after manual installation of SnpEff, <code>vcf2rdf</code> , and Blazegraph. Source code only (https://github.com/MU-Data-Science/GAF).	●	●	○	●	●	○	●
AgroLD VCF2RDF [14] Named in [14] among the AgroLD conversion pipelines (GFF2RDF, GAF2RDF, VCF2RDF) at https://github.com/SouthGreenPlatform/AgroLD_ETL . AgroLD itself is ontology-based, but no VCF vocabulary, usage, or output is documented. Source code only.	●	○	○	○	○	○	●
<i>Virtual semantic access</i>							
Semantic Beacons [15] Maps Beacon API responses, not VCF files, to RDF on the fly with RML over JSONPath, served as Triple Pattern Fragments and queried with Comunica. Its SO, FALDO, and GENO schema covers variant positions and alleles, which the authors describe as partial coverage of the Beacon specification; Beacon responses carry no header or per-sample genotypes. Implementation details are deferred to https://github.com/SemanticBeacon/SemanticBeacon .	●	○	●	○	●	○	●

S2 Shared-input converter comparison

Table S1 records what each converter documents. This section reports what four of them write when they convert the same inputs as VCF-RDFizer, measured with the paper’s existing measurements: the validator’s content questions Q1–Q8 against the oracle, conversion wall time and peak resident memory, triples per record, and gzipped N-Triples size (main Figure 4). The run is experiment 18 of `vcf-rdfizer-testing` at commit `e3743590`, on `vcf-bench-1` on 8 October 2026, with three replicates of every conversion; its records are

archived in `benchmark-results/vcf-bench-1/18_converter_comparison/`. The whole experiment runs from one Docker Compose file, so the host needs only Docker: `CONV_DATA=... benchmarks/converters/run.sh`.

S2.1 Converters, builds, and options

Each converter ran in its upstream release image wherever upstream publishes one, and with the options that make its output most complete rather than with its defaults (Table S2). Images built for the experiment start from a Debian base pinned by digest and take their packages from `snapshot.debian.org` at a fixed date; every downloaded artifact was checked against a pinned SHA-256, and the run records the resolved Compose file and every image ID. Converter containers ran without network access, as the calling user. A static BusyBox `time -v`, mounted read-only into each container, measured wall time and peak resident set size, the quantities VCF-RDFizer’s own run summary reports for its stages; VCF-RDFizer’s wall time spans all its stages, including starting a container for each. Apache Jena `riot` 6.2.0 streamed each native output to N-Triples, which were counted, gzipped, and indexed with the QLever build in the VCF-RDFizer image.

Table S2. Converters in the shared-input comparison: the version run, how it was obtained, and the options used.

Converter	Version	Obtained as	Options; output
VCF-RDFizer	v3.3.1 (b25fb7b)	Release image <code>sha256:3ad71b1a...2993</code> , driven by the release’s command-line tool	Expanded samples, space-optimized storage, N-Triples only; gzip-framed N-Triples
JVarkit <code>vcf2rdf</code> [2]	Commit <code>fc2356a3</code>	Built from source, with its git submodule; no jar is published, and the release tag was moved after publication, so the commit is the pin. The SHA-256 of every library jar the build fetched is recorded	<code>--hide ''</code> (the default hides ALT, FILTER, and genotypes); Turtle
TogoVar <code>vcf2rdf</code> [1]	v1.0.0-beta.12 (f17835a6)	Upstream Docker Hub image, by digest	<code>generate config</code> for the input’s assembly, then <code>convert --no-normalize</code> , which keeps POS, REF, and ALT as written; Turtle
SPARQLing Genomics <code>vcf2rdf</code> [4]	0.99.11 (1859f6f8)	Upstream release image (a Guix pack), loaded from its release tarball after a checksum check	Defaults, which already write INFO, FORMAT, and the header; N-Triples
BioInterchange [9]	2.0.5 (49d57012)	The brewsci Homebrew bottle, still served by its registry and addressed by its SHA-256. The source build no longer works, because a dependency’s repository has been deleted	Defaults; line-delimited JSON-LD, whose contexts now return HTTP 404 and were inlined from the pinned commit

The other entries in Table S1 were not run. The VCF2RDF web service is offline and documents no command line; the BioHackathon `vcf2rdf` gem was never published; `rdf-vcf` has been unmaintained since 2015 and names no vocabulary; `bio-vcf` writes RDF only through user templates without a documented vocabulary; VariantKG converts with SPARQLing Genomics’ `vcf2rdf`; AgrolD documents no VCF conversion; and Semantic Beacons maps Beacon responses rather than VCF files.

S2.2 Questions and ports

Every graph was compared by the same script, inside the VCF-RDFizer image, with that image’s validator: the same oracle over the source VCF, QLever build, result normalization, and comparator. Only the SPARQL differs: each converter’s Q1–Q8 are ported to its vocabulary, and VCF-RDFizer’s are its validator’s own queries. A question a graph cannot answer has no port; its reason, with evidence from the converter’s source, is recorded instead and reported as not represented, and the script refuses to run when a question has neither. Q9–Q13 were excluded: Q9 and Q10 count VCF Core predicates and classes, and Q11–Q13 hash VCF-RDFizer’s record IRIs.

Ports read only what a graph states. Where a converter writes two VCF values the same way, the port reads that state as the value the converter’s code derives it from, and the comparison shows whether the conflation costs anything on these inputs: Jvarkit writes no FILTER triple for either PASS or ‘.’ (read as PASS), writes no ploidy (haploid calls read as homozygous), and types neither a no-call nor a record without GT; TogoVar replaces CHROM with a reference IRI from its configuration, which the Q1 port maps back through the same configuration; SPARQLing Genomics writes no record node, so its calls are regrouped through the record index in their IRIs, as its manual documents, and it writes only the first ALT allele; and BioInterchange writes a missing allele as REF, writes ‘*’ as an empty string, and strips the VCFv prefix from `##fileformat`. TogoVar and SPARQLing Genomics lose the order of multiple FILTER codes.

S2.3 Results

Every conversion and comparison completed (52 recorded steps, all with exit status 0). VCF-RDFizer answered Q1–Q8 as the oracle did on both inputs; the other graphs’ differences come from what they hold:

- **Jvarkit.** Its record IRI is `urn:variant/CHROM:POS:REF`, so the five pairs of 1000 Genomes lines that share CHROM, POS, and REF (for example `20:283049 C>CGAT` and `C>T`, both with ID ‘.’) merge: 9,995 records instead of 10,000, five more multiallelic records, and every later count shifted. On the HG005 slice it answered Q1–Q6 as the oracle did.
- **TogoVar.** It drops alleles outside `[ACGTURYKMSWBDHVN]`: the 13 symbolic records of the 1000 Genomes slice disappear, and `chr1:1663386 C>*,T` on the HG005 slice becomes an SNV (Q2, Q3). The same five pairs merge, because TogoVar writes no record node. It writes no samples or header lines, by design.
- **SPARQLing Genomics.** A call is written only when it carries an alternate allele, so the 1000 Genomes graph holds the 1,401 of 10,000 records with a carrier, no homozygous-reference calls, and allele numbers over carriers only (Q6); multiallelic records are recognized only through an allele index of 2 or more, 19 of 70. On the HG005 slice, where every call carries an alternate allele, it answered Q1–Q7 as the oracle did; htlib adds a PASS FILTER definition the file does not declare (Q8).
- **BioInterchange.** Its header document for the HG005 slice (line 2 of 100,003) is not valid JSON, because the tool writes strings without escaping them, so the header is absent from that graph (Q7, Q8). All 100,001 record documents parse, and on the 1000 Genomes slice it answered all eight questions as the oracle did.

Replicates. Each converter’s three outputs were byte-identical, with two exceptions. VCF-RDFizer’s gzip framing differed between replicates, but its sorted triples were identical in all three replicates of both inputs; on the HG005 slice they equal those of the v3.1.0 conversion (Table S13).

Table S3. Conversion cost and output size in the shared-input comparison: median wall time and largest peak resident set size over three replicates, and the size of replicate 1’s output once streamed to N-Triples. QLever index times are for the same graphs.

Converter	Input	Wall (s)	Peak RSS (GB)	Triples	Triples per record	N-Triples.gz ÷ VCF.gz	QLever index (s)
VCF-RDFizer	HG005	91.6	1.85	17,098,746	171.0	25.5	20.8
	1000 Genomes	31.4	0.93	5,359,578	536.0	136.9	7.0
JVarkit	HG005	2.2	0.43	1,401,613	14.0	2.1	1.8
	1000 Genomes	1.4	0.30	1,020,454	102.0	18.2	1.4
TogoVar	HG005	1.3	0.01	5,728,120	57.3	15.4	6.8
	1000 Genomes	0.1	0.01	564,940	56.5	35.0	1.0
SPARQLing Genomics	HG005	6.3	0.01	2,958,769	29.6	4.8	3.3
	1000 Genomes	0.4	0.01	180,053	18.0	4.1	0.5
BioInterchange	HG005	1.5	0.02	4,559,438	45.6	17.0	6.6
	1000 Genomes	0.4	0.01	2,031,215	203.1	154.3	3.2

BioInterchange’s replicates differed only in their **Context** and **Summary** documents, which record run paths and timestamps.

IRI warnings. riot parsed every output, warning about three converters’ IRIs: VCF-RDFizer’s record IRIs, `file://<name>.vcf#record/n`, place the file name where a URL’s host would be (an “unwise” IRI; 23.6 million warnings on the HG005 graph); SPARQLing Genomics’ `origin://<md5>@<k>` uses deprecated user information (3.1 million); and JVarkit’s `urn:variant/...` has an invalid URN namespace (0.6 million). None prevented parsing or indexing; TogoVar’s and BioInterchange’s outputs drew no warnings.

S3 Representation and implementation details

S3.1 Model and sample profiles

VCF-RDFizer uses the VCF Core vocabulary (`vcfc:`, <https://w3id.org/vcf-core/vocab#>) to represent file and header assertions, variant records, and sample/genotype information [16]. The model aligns selected terms with FALDO, the Sequence Ontology, HERO, and GA4GH VRS [17, 18, 19, 20]. SHACL supplies value and cardinality constraints rather than OWL restrictions. Figure S1 shows the emitted backbone, and Table S4 separates implemented support from completed assessment.

Expanded output represents each sample call and FORMAT value as an addressable resource. Condensed output stores corresponding values in sample-ordered vectors. Both profiles retain the same file, header, and record context. For V records, S samples, and F FORMAT keys in a uniform layout, the sample-layer structure scales as

$$\begin{aligned} T_{\text{expanded}} &= \mathcal{O}(S + VSF), \\ T_{\text{condensed}} &= \mathcal{O}(S + VF). \end{aligned} \tag{1}$$

Condensed output still stores all VSF values inside literals, so bytes increase with sample count even when repeated triple structure does not. Deterministic IRIs identify observations by source file and record position, while linksets add cross-file identity separately. GA4GH VRS computed identifiers could complement the evaluated SPDI route but were not implemented [20].

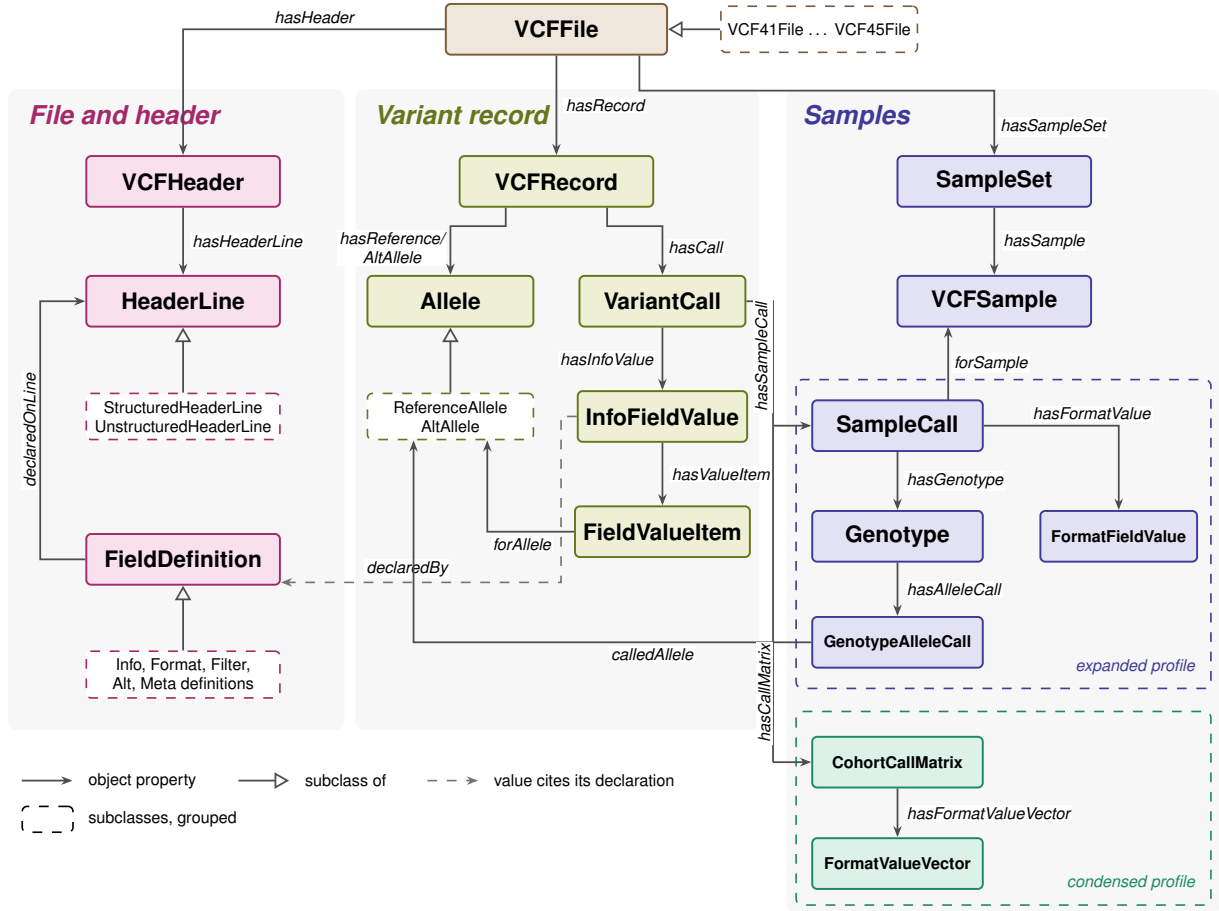


Figure S1. Simplified VCF Core model emitted by the evaluated converter. A versioned file connects header declarations, records, and sample information, while values retain links to their interpretation context. Expanded output exposes sample calls, genotypes, and allele calls, whereas condensed output uses sample-ordered vectors. The complete vocabulary is documented at <https://ecrum19.github.io/vcf-core-vocabulary/>.

S3.2 Conversion and incremental representation construction

Conversion derives record, header-line, and file-metadata TSV sources because the implemented RML path does not read VCF directly. RMLStreamer applies Turtle-encoded mappings to the record backbone, while direct emitters add samples and values requiring row-specific datatype or structure handling [21, 22]. RML contributes nine triples per record, or 5.3% of the HG005 graph, and emitters contribute the remaining 94.7%. The percentages describe the evaluated hybrid implementation rather than a general limitation of RML.

Each part is appended to plain N-Triples or one member of a concatenated-gzip stream and then deleted. Standard gzip readers expose the members as one N-Triples sequence. HDT and COTTAS construction, RDF compression, and transport packaging are independent options. The representation builders consume one complete-line chunk at a time, so the process avoids a second full decompressed aggregate. HDT parts are merged pairwise with `hdtc`, while COTTAS parts are merged in one streaming k -way pass [23, 24, 25, 26, 27]. Default chunk targets are 512 MiB, bounded between 128 MiB and 1 GiB; six feasibility runs used smaller chunks. Before raw RDF is removed, each artifact is decoded with its native reader and its triple count compared with the source graph. The check covers decoding and count preservation, not complete RDF-term equality. QLever decodes an input artifact and builds its own index, so QLever timings do not measure native HDT or COTTAS querying. The native COTTAS route uses a DuckDB-backed Comunica engine over Parquet [28].

Table S4. Implemented capabilities and assessed scope. Each status gives release support, then the completed evaluation in italics; unprefixed section references are to the main text.

Capability	Status	Notes and limits
Core fields (CHROM, POS, ID, REF, ALT, FILTER)	Supported (v3.1.0) <i>Validated</i>	Q1–Q4 equal in completed comparisons and in the consumer WGS validation run (Section S4.2)
QUAL	Supported <i>Q11 record digest</i>	Typed literal for every record; a missing QUAL is <code>vcfc:Null</code>
INFO	Supported; structured by default <i>Q11, Q12 digests</i>	Typed value per record and key, per-allele items for Number=A/R/G, plus the complete <code>infoRaw</code> string
FORMAT, sample, and genotype values	Supported <i>Q5–Q6 exact; Q13 digest</i>	Half-calls, haploid calls, and ploidy 4/3 survive conversion (Table S15); called allele, sample index, and phasing require the deeper shapes for the tested mutations
Header metadata	Supported <i>Census queries and shapes</i>	Header attribute–value agreement requires the deeper shapes, not merely the default profile
Multi-sample conversion	Supported <i>Measured, 1–2,504 samples</i>	Condensed adds 5 triples per sample; expanded adds 25 per sample per record (627M triples for 10,000 records at 2,504 samples), so the whole cohort was converted in neither (Section 4.3)
Structural and symbolic alleles	Lexically supported <i>HGSVC2 converted; Q2 passes on symbolic and breakend fixtures</i>	ALT remains the source lexical field; allele-value mutations are caught only by the full shape profiles
Queryable HDT, COTTAS	Supported <i>Measured</i>	118 HDT and 34 COTTAS artifacts, all decode checks passed; COTTAS smallest throughout (Section 4.3)
Gzip/Brotli packaging	Supported <i>Covering set (Section S10)</i>	Packages must be unpacked before querying; no codec comparison
Policy annotations	Plug-in v0.2.0 (v3.3.0) <i>Real VCFs, up to a complete single-sample VCF</i>	ODRL on files and declared graph selections, DUO purposes; every view checked against a VCF-side oracle; simulated consents, single-sample files; governed release, not anonymization (Section 4.2)
External linking	SPDI, Ensembl-gene, and MyVariant linkers (v3.3.0) <i>Real VCFs</i>	SPDI joins eligible alleles under the stated reference and normalization assumptions; tier 3 confirmed 92–93% of rsID rewrites (Section S5)

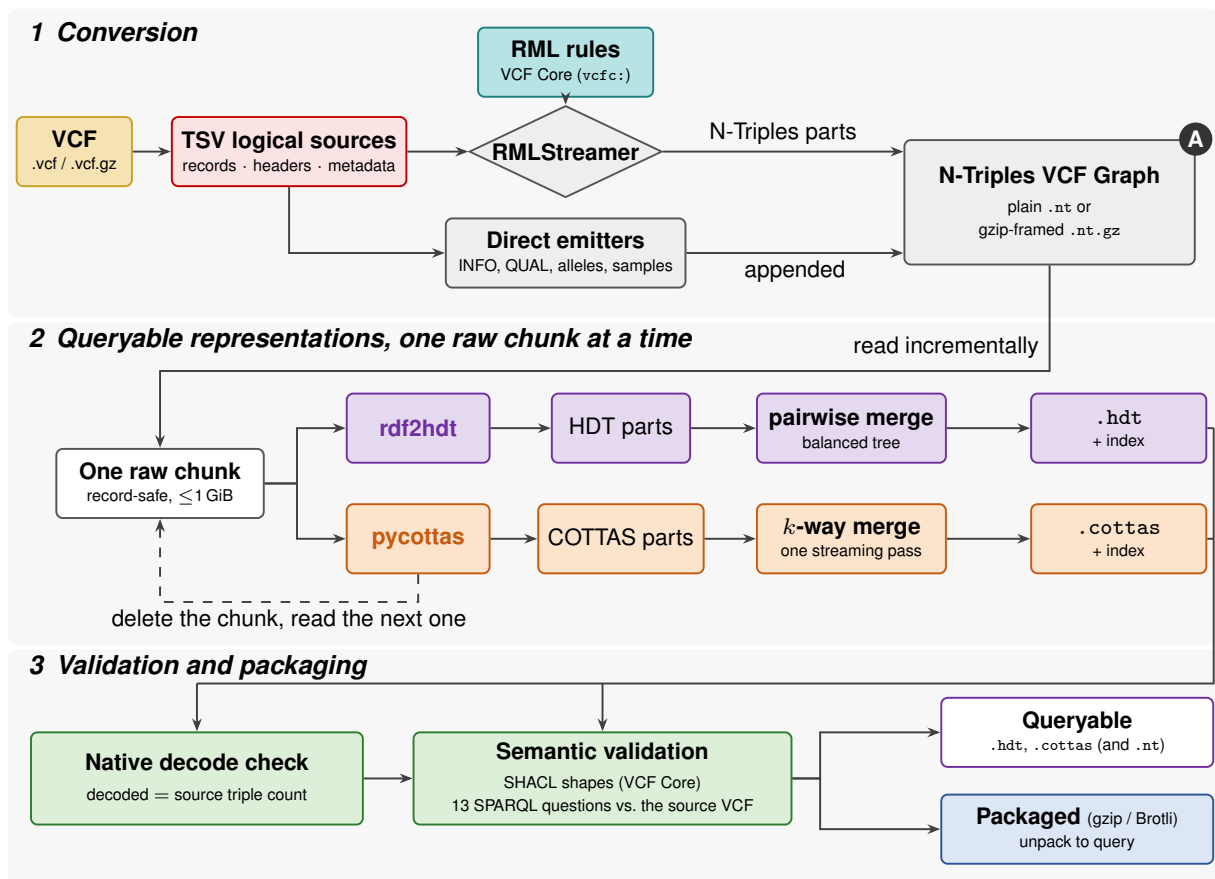


Figure S2. VCF-RDFizer v3.1.0 implementation. RML mappings and direct emitters produce the source graph, which is appended incrementally as plain or gzip-framed N-Triples. Optional HDT and COTTAS builders consume one record-safe chunk at a time before merging and indexing. Native decoding, source-comparison queries, and enabled SHACL profiles assess different properties.

S4 Validation specification, scope, and sensitivity

S4.1 Queries, normalization, and engines

Validation compares thirteen SPARQL results with answers derived from the source VCF using `cyvcf2` and exact FILTER text from `bcftools query` [29, 30, 31]. Fourteen preflight queries first inspect syntax-independent graph properties, including missing values, POS typing, core-property cardinality, sample representation, inventories, and triple counts. Table S5 defines the source-comparison questions.

Table S5. The thirteen source-comparison queries. Each normalized SPARQL result must equal its parser-derived counterpart; Q4 uses exact `bcftools FILTER` text. Ti/Tv, call rate, and allele frequency are derived after their counts agree. Q11–Q13 compare hash-bucket summaries; Q9, Q10, Q12, and Q13 depend on the selected representations.

Query	Result compared, fields, and scope
<i>Record level: every input</i>	
Q1 Record density	Records per contig and 1-Mb window (CHROM, POS); checks coordinates and integer positions
Q2 Allele shapes	Records per class: SNV, multiallelic, insertion, deletion, substitution, symbolic/breakend, no ALT (REF, ALT)
Q3 Ti/Tv	Transitions, transversions, and biallelic SNVs; single-base A/C/G/T alleles only
Q4 FILTER values	Records per exact FILTER string: PASS, . (not applied), or failed codes, combined codes kept whole
<i>Genotype level: inputs with sample columns and GT</i>	
Q5 Genotypes	Calls per sample and genotype class, including missing, haploid, and other-ploidy calls; phased and unphased alike
Q6 Allele counts	Sites per recomputed (AC, AN) pair; single-ALT sites with complete 0/1 haploid or diploid calls
<i>Metadata and graph structure</i>	
Q7 File metadata	Exact <code>fileformat</code> , <code>reference</code> , and <code>source</code> strings, with markers for absent declarations
Q8 Header census	## lines per header key (INFO, FORMAT, FILTER, contig, ...)
Q9 Predicate census	Triples per predicate against the counts expected from the VCF
Q10 Class census	Resources per <code>rdf:type</code> against the expected inventory
<i>Value preservation: IRI-bound digests</i>	
Q11 Record digest	Records per bucket over each record’s IRI, CHROM, POS, ID, REF, ALT, QUAL, FILTER, and raw INFO
Q12 INFO digest	Structured INFO values per bucket, bound to record and key; empty in raw-only INFO mode
Q13 FORMAT digest	FORMAT values (expanded) or sample-ordered vectors (condensed) per bucket, beyond GT

Normalized results retain contig and sample identifiers, distinguish PASS from ., and report missing, extra, or differing rows. Q5–Q6 apply only to inputs with sample data and GT. Q11–Q13 bind values to record IRIs and count them in 256 buckets derived from SHA-256. The digest summaries extend value coverage but can miss changes that preserve the bucket histogram.

The same SPARQL text runs under QLever, Comunica over N-Triples, Comunica over HDT, and the DuckDB-backed COTTAS engine [32, 33, 34, 35, 28]. The default SHACL profile checks cardinality and datatypes, while two deeper profiles check uniqueness and value agreement. A passing query or shape result establishes only the specified summary or enabled constraint.

S4.2 Completed comparisons and the consumer WGS validation run

The base campaign attempted 78 validations and completed 76. The completed runs yielded 984 equal applicable query comparisons and four not-applicable genotype comparisons on zero-

record artifacts. Paired validation reached 17.1 million triples, while the two malformed sites-only validations did not complete. The default shapes ran in 58 validations up to 0.96 million triples with no violations. Repeatability checks produced identical sorted triples, preserved digests through packaging, and left artifacts unchanged after re-indexing.

Table S6. Validation results by assessment layer. Counts report completed checks; fault sensitivity applies only to the defined mutation set and enabled profiles.

Check	Passed	What it establishes
<i>Is the graph faithful to the source VCF?</i>		
N-Triples syntax (rapper)	76/76	The serialization parses
Paired queries vs cyvcf2 (Q1–Q13)	984/984 [†]	Exact equality of normalized query results
Cross-check invariants	1,144/1,144	E.g. transitions + transversions = biallelic SNVs
Cross-engine agreement	25/25	Four-engine validations; 51 other completions used one engine
Native decode and triple count	152/152	118 HDT, 34 COTTAS; source = decoded
<i>Is the pipeline reproducible?</i>		
Determinism, two runs	1/1	Identical SHA-256; 17,098,746 triples, 0 duplicate lines
Compress and decompress	1/1	Lossless through the packaging path
Index idempotence	1/1	Re-indexing does not perturb the artifact
<i>Would a fault actually be noticed?</i>		
Faults caught, queries only	96/113 (0.850)	17 faults, spanning 10 mutation classes, are invisible to the queries
with the default shape profile	96/113 (0.850)	Cardinality and datatypes catch none of the 17
with all three shape profiles	113/113 (1.000)	Uniqueness and value agreement catch all 17; both controls still pass
SHACL shapes, default profile	58/58	Cardinality and datatype constraints, 0 violations

[†]Four further comparisons were verified as not applicable and are excluded from the 984 applicable comparisons.

The complete HG005 retrieval run compared all thirteen answers at 657,425,805 triples without SHACL and is not included in the 984-comparison denominator. The consumer WGS validation run converted 250,000 NG131FQA1I records into 58,231,176 triples. A v3.1.0 run first reported three discrepancies because the oracle omitted emitted phase sets and compared non-canonical decimal QUAL strings. Direct field comparison found no conversion differences, and canonicalizing QUAL reproduced the graph-side digest. Version 3.3.1 corrected the oracle, added expectations for additional emitted structures, and validated record-local default shapes in batches. The rerun with the published v3.3.1 image agreed on all thirteen results and reported no violations across 117 batches, with 46 non-blocking recommendations for missing VCF 4.5 **Source** and **Version** attributes. The batched run took 91 minutes with four workers, and available memory never fell below 19.5 GB.

S4.3 Fault sensitivity and oracle limits

The fault set contains 113 targeted changes to coordinates, alleles, filters, metadata, phasing, sample indexes, and relationships. Source-comparison queries detected 96 faults. The default shapes detected none of the remaining 17, while all three profiles together detected every fault and accepted both unmutated controls. The 17 additional detections span ten classes, including allele links, header attributes, record and sample indexes, called alleles, and phasing. The deeper profiles took 33.7 and 73.2 seconds on approximately 2,000 triples, compared with 1.7 seconds for the default profile. The result therefore describes the defined fault set on a small graph rather than an estimated detection rate for arbitrary data.

Version 3.3.1 applies record-local default shapes in batches at any graph size, while deeper cross-record profiles retain a whole-graph limit. The parser and graph routes read different representations but share selected definitions and representation expectations. An incorrect dependency

can therefore affect both the converter and its oracle, as occurred for the missing-allele defect. Text comparisons, source-derived summaries, structural constraints, and fault injection provide complementary evidence rather than an infallible oracle. Local-allele semantics remain outside the completed oracle coverage.

S5 Linked genomic workflow: protocol, policies, and detailed results

S5.1 Scope, linking, and policy evaluation

The linked workflow tests relationships shared by the three motivating applications rather than implementing complete clinical or governance systems. For reanalysis, the workflow joins observations to fixed ClinVar and population-frequency releases. For pharmacogenomics, the workflow tests cross-resource linking but not diplotype interpretation, medication orders, or prescribing guidance. For consent-aware reuse, the workflow evaluates simulated purposes, withdrawals, and record selections but not real consent adjudication or authenticated access. Figure S3 summarizes the implemented path.

Linking. `vcf-rdfizer-link` executes Turtle-declared linkers and writes a separate N-Triples linkset with coverage and skipped-record reports. Tier 1 rewrites existing identifiers, Tier 2 joins digest- and assembly-pinned references, and Tier 3 confirms identifiers through a rate-limited live service with cached responses. The evaluated SPDI linker minimally trims explicit alleles and harmonizes contigs to RefSeq accessions, while the gene linker records Ensembl 116 genes overlapped by the REF span [36]. Minimal trimming supports the tested joins but does not establish canonical identity in every repeat context. The live MyVariant.info linker confirms rsIDs and fails without publishing a partial linkset when its service fails [37].

Policy evaluation. `vcf-rdfizer-policy` evaluates ODRL rules over files or declared graph selections [38]. The VCF Core policy profile applies default-deny and deny-overrides semantics and uses a subset of DUO purposes by default [39]. Ownership rules remove dependent calls, alleles, sample calls, and headers with a withheld source resource. Each release view is checked against the policy digest, permitted and prohibited membership, dangling references, and a record oracle derived from the VCF text. Released genotypes retain source IRIs and are not anonymized, and obligations are recorded but not enforced. Selector parameters are placed inside the outer SPARQL group; a prior trailing placement produced an empty region selection and was corrected before the reported experiments.

S5.2 Application question, comparison workflows, and evaluation arms

The application asks which observed variants in the fixed ACMG SF v3.2 81-gene set have a ClinVar classification, which participant and gene belong to each match, and which matches each requester may receive [40]. ClinVar records require at least one review star. Pathogenic and likely pathogenic matches are reported separately but are not treated as reportable secondary findings.

The RDF workflow links converted VCFs to ClinVar and gene resources, materializes each requester view, and queries the view with SPARQL against QLever [32]. The conventional workflow harmonizes contigs, annotates `CLNSIG`, `CLNREVSTAT`, and `GENEINFO` with `bcftools`, and applies the same case definition and release rules in Python [30]. Sorted tuples of participant, gene, contig, position, and alleles should agree before timings are reported. The routes share preprocessing and the case definition but use separate downstream implementations.

1. Prepare each file and add explicit links

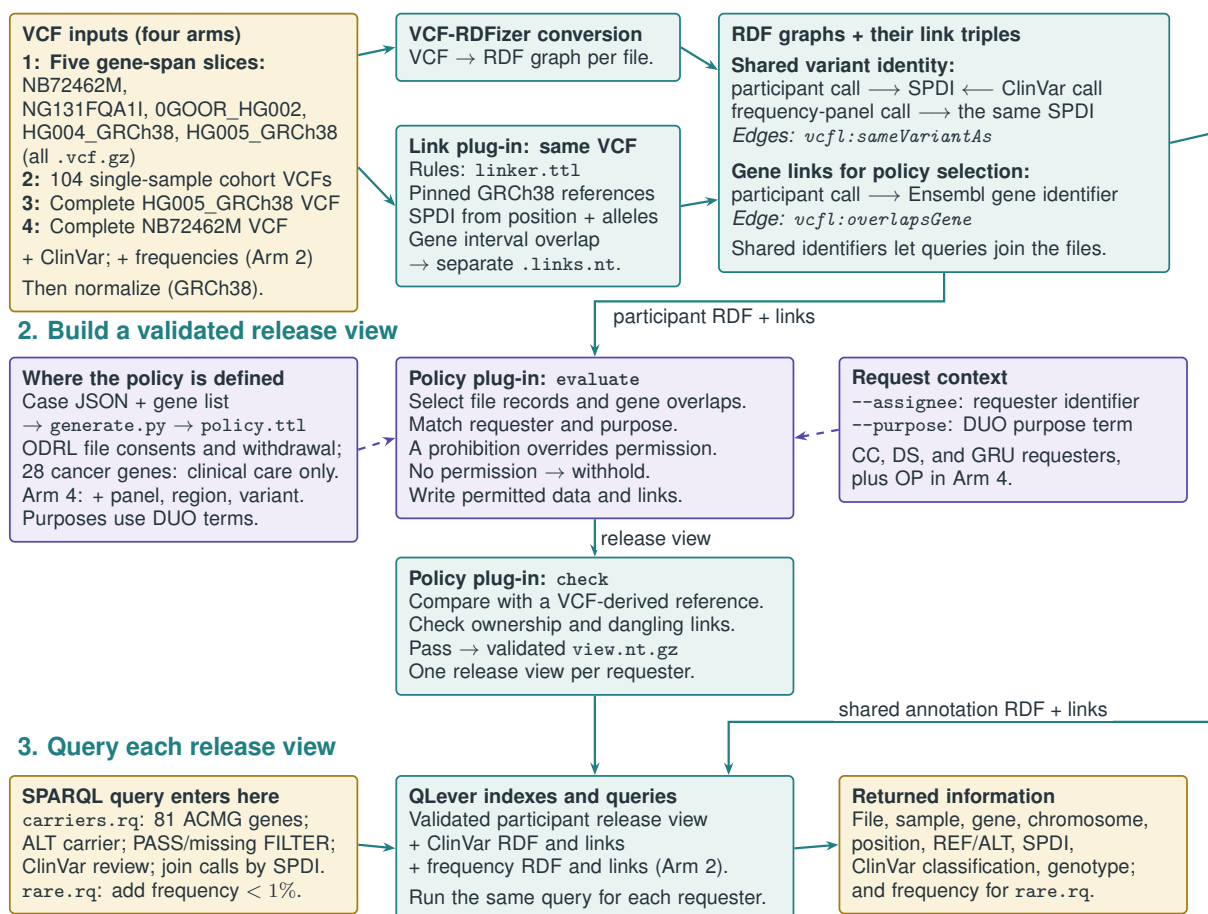


Figure S3. The implemented path from VCF files to a permitted answer. The diagram follows the use-case runner, policy generator, linker definitions, and SPARQL files. Linkers read the normalized VCFs directly, and conversion output and links are stored separately; inputs are restricted to the ACMG gene spans except in Arms 3 and 4. Policies are enforced when each release view is materialized: requester identity and purpose are evaluation inputs, while the SPARQL query holds the analysis filters. The cancer-gene policy uses Ensembl overlap links, whereas the query reports the ACMG genes named in ClinVar’s annotation. Requesters are clinical care (CC), disease-specific cardiovascular research (DS), general research use (GRU), and, in Arm 4, care by the participant’s own physician (OP).

Arm 1 contains ACMG gene-span slices from five single-sample VCFs, Arm 2 contains 104 high-coverage 1000 Genomes participants, Arm 3 contains the complete HG005 VCF, and Arm 4 contains the complete NB72462M VCF [41, 42, 43]. Arms 1 and 2 use normalized and split records within 7.1 Mb of Ensembl 116 gene spans. Arm 2 removes repeated panel-level INFO and stores the required frequencies once in a sites-only graph. All arms use the 2026-09-13 ClinVar release [44]. Simulated release rules combine file permissions, four seeded withdrawals in Arm 2, a 28-gene cancer restriction, and requester purposes for clinical care, cardiovascular research, and general research. Arm 4 adds care by the participant’s physician and combines a 40-gene cardiovascular panel, an APOE region, and one DSP variant so that every requester receives a non-empty subset of the same complete VCF. No arm tests selective masking of samples within a multi-sample VCF.

Four applied changes test a withdrawal, a new purpose, replacement of the cancer panel, and an HFE C282Y restriction. After each change, both workflows should release the same fixture records. Line counts describe the two implementations and do not measure usability or maintainability.

S5.3 Linking results, matches, and rule changes

The SPDI linker emitted links for eligible explicit alleles under the stated assembly and normalization assumptions. A seven-variant BRCA1 check agreed with NCBI expressions outside repeats, while repeat-region spans differed from NCBI contextual representations. The live tier confirmed 9,223 of 9,951 and 9,479 of 10,316 PGP rsID rewrites. Its 21 batch requests reached MyVariant.info once, on 2026-09-28; v3.3.1 confirmed the same links offline from the recorded responses, which the linker keys by request (Section S6.2). The Ensembl live variation service returned HTTP 500 after three retries, and no partial linkset was published.

Table S7. Calls linked per linker, summed over each arm’s files. SPDI eligibility requires explicit alleles. Every arm used v3.3.1, whose gene linker also links spanning-deletion (*) records, which the SPDI linker skips: 65 in Arm 1 and 15,032 in Arm 4. A further 19 HG002 calls in Arm 1 overlap genes but are ineligible for the SPDI linker. The gene linker links no symbolic allele or breakend.

Linker	Tier	Arm 1	Arm 2	Arm 3	Arm 4	ClinVar	Result
spdi	2	51,302	1,149,198	3,887,200	4,991,223	331,413	Every eligible call; joins the input VCFs to ClinVar across contig names; 89s for the complete HG005 VCF
ensembl-genes-grch38	2	51,386	1,149,198	1,704,954	2,063,627	—	Reloads the 108 MB GFF3 per file: 19 of Arm 2’s 22 min
rsid-dbsnp	1	20,267	—	—	4,974,526	—	PGP VCFs’ rsIDs, rewritten unchecked
rsid-myvariant	3	18,702	—	—	—	—	92–93% of tier-1 links confirmed in 21 requests; the three unconfirmed rsIDs checked were merged
rsid-ensembl	3	0	—	—	—	—	Persistent HTTP 500; failed closed, no partial linkset

The RDF and conventional workflows returned identical requester-specific match sets in every arm. Every release view passed the policy and source checks. The complete HG005 result equaled its gene-span result, and the Arm 2 frequency join agreed with the conventional position-and-allele lookup.

Table S8. Requester-specific participant–variant–gene matches; the RDF and conventional workflows agree in every cell. All: no policy; OP: care by the participant’s own physician (Arm 4 only); CC: clinical care; DS: cardiovascular disease-specific research; GRU: general research use. Triple counts exclude links; the rare-variant row also joins panel frequencies at AF < 0.01.

Arm	Files	Records	Triples	Matches by requester				
				All	OP	CC	DS	GRU
1: five gene-span slices	5	51,386	9.0M	7,211	—	2,878	2,988	1,987
2: cohort	104	1,150,288	67.8M	173,102	—	63,529	92,870	48,207
rare in the panel				4,154	—	1,621	2,198	1,144
3: complete HG005	1	3,887,810	662.4M	1,496	—	1,496	0	0
4: complete NB72462M	1	5,063,417	1,209.7M	1,382	1,382	1,382	722	983

Arms 1, 3, and 4 also agreed on the number of released records for each requester; Arm 2 did not, as described below. In Arm 4, the participant’s physician received the complete VCF, clinical care excluded six APOE records, disease-specific research retained only the cardiovascular panel, and general research excluded cancer-gene, APOE, and single-variant selections.

The record comparison exposed gene-linking gaps that match comparison and release validation did not detect. Version 3.3.0 skipped spanning-deletion records, so gene-based rules could

Table S9. Arm 4: what each rule withheld from the complete NB72462M VCF (5,063,417 records after normalization), per requester. The simulated consent permits clinical care and general research use. A record withheld by several rules is counted once, under the first; for DS, APOE and the DSP variant fall outside its panel. OP: the participant’s own physician; other abbreviations as in Table S8.

Rule	Selects records by	OP	CC	DS	GRU
Cancer genes (28): CC only	gene links	—	—	2,513	2,513
Cardiovascular panel (40): all DS sees	complement of the panel’s gene links	—	—	5,054,507	—
APOE: physician only	coordinates on chr19	—	6	—	6
DSP variant: not GRU	SPDI identity	—	—	—	1
Records released		5,063,417	5,063,411	6,397	5,060,897
Matches		1,382	1,382	722	983

not select them. Version 3.3.1 links spanning deletions by REF span, and Arms 1, 3, and 4 agree with the conventional workflow record for record. Symbolic alleles and breakends remain outside the gene linker because their extent requires additional interpretation. In Arm 2, the RDF views therefore released 110 records to DS and 58 to GRU that the conventional workflow withheld: every one is a symbolic structural variant (<INS:ME: . . .>, <DUP>, or <INS>) in a restricted cancer-predisposition gene, such as *BRCA2*, *NF2*, or *STK11*. No match changed, because the case definition joins explicit alleles only, and each view still passed its own checks, which share the linker’s gene links.

Restricting HFE C282Y required a policy edit without changing the RDF policy engine and selected the HG002 record despite its unprefixed contig name. The conventional workflow required a small code and data edit. The other three changes were similarly sized data edits in both routes.

Table S10. Authored content and rule changes in Arm 1, in non-blank, non-comment lines. After each change, both workflows returned the same requester-specific records. POA: DUO’s population-origins-or-ancestry purpose, outside the GRU permission used here.

	RDF workflow	Conventional workflow
<i>Initial authoring</i>		
Rules written	72: policy 41, query 31	131: shell 33, Python 98
Data written	28 gene IRIs; the query’s gene list (1 line)	Consent and requester tables (100 lines); gene list (81)
Linkers, selectors, and profile	Shipped with the tool	—
<i>Rule changes</i>		
A participant withdraws	Policy +2/ − 1	Data +2/ − 1
A requester with a new purpose (POA)	Purpose vocabulary +2	Data +4
The panel becomes 40 cardiac genes	Policy +40/ − 28 (gene IRIs)	Data +41/ − 29 (gene symbols)
One variant (HFE C282Y), clinical care only	Policy +6, no engine change	Code +3, data +9/ − 1

S5.4 Stage costs

Conversion and linking are paid once per arm, while view materialization, validation, indexing, and querying are repeated per requester. Query times ranged from 146 to 217 seconds across the four arms, consistent with a dominant fixed ClinVar join rather than generally scale-independent retrieval. Arm 2 release writing took 59–81 seconds per requester; pre-release builds took 20–27 minutes before binding selections were combined, filtering was parallelized, and term ancestry was cached, with byte-identical views. Arm 3 peaked at 11.2 GB of host memory in use and 31 GB of disk. For Arm 4, conversion required 92.7 minutes, linking 23.1 minutes, and each selective view 55–74 minutes to write and validate.

Table S11. Stage costs of the linked workflow for every arm and requester; the data behind Figure S4. Conversion and linking are paid once per arm, and both include ClinVar. Write is the policy evaluation that streams the release view. Check is release validation, measured as the interval before the next requester’s evaluation, so it is not recorded for the last requester; for a non-empty view it includes building the view’s own index. Index is the build of the query index, and query the median of three runs over it. Records are those in the release view; triples are the query index’s, including ClinVar and links. Abbreviations as in Table S8.

Requester	Records	Index triples	Write <i>min</i>	Check <i>min</i>	Index <i>min</i>	Query <i>s</i>
<i>Arm 1: five gene-span slices: convert 4.2 min, link 1.5 min</i>						
All (no policy)	51,386	42,543,789	—	—	2.3	177
CC	19,625	37,514,230	0.15	0.50	2.2	176
DS	23,933	37,813,325	0.13	0.53	2.2	171
GRU	16,105	35,967,049	0.13	—	2.2	173
<i>Arm 2: cohort of 104: convert 34.1 min, link 22.2 min</i>						
All (no policy)	1,150,288	110,268,925	—	—	3.2	203
CC	421,991	65,787,093	0.98	3.0	2.5	180
DS	685,928	81,878,625	1.4	4.9	3.1	188
GRU	356,835	61,794,418	1.2	—	2.4	179
<i>Arm 3: complete HG005 VCF: convert 55.3 min, link 8.1 min</i>						
All (no policy)	3,887,810	701,489,559	—	—	45.9	217
CC	3,887,810	701,489,559	38.9	70.0	46.5	217
DS	0	33,428,173	28.5	1.1	2.3	0.07
GRU	0	33,428,173	28.7	—	2.3	0.07
<i>Arm 4: complete NB72462M VCF: convert 92.7 min, link 23.1 min</i>						
All (no policy)	5,063,417	1,255,334,983	—	—	83.2	184
OP	5,063,417	1,255,334,983	72.8	117.0	84.0	182
CC	5,063,411	1,255,333,543	74.3	116.6	83.8	181
DS	6,397	34,971,594	55.1	9.7	2.2	146
GRU	5,060,897	1,254,725,280	73.4	—	83.5	181

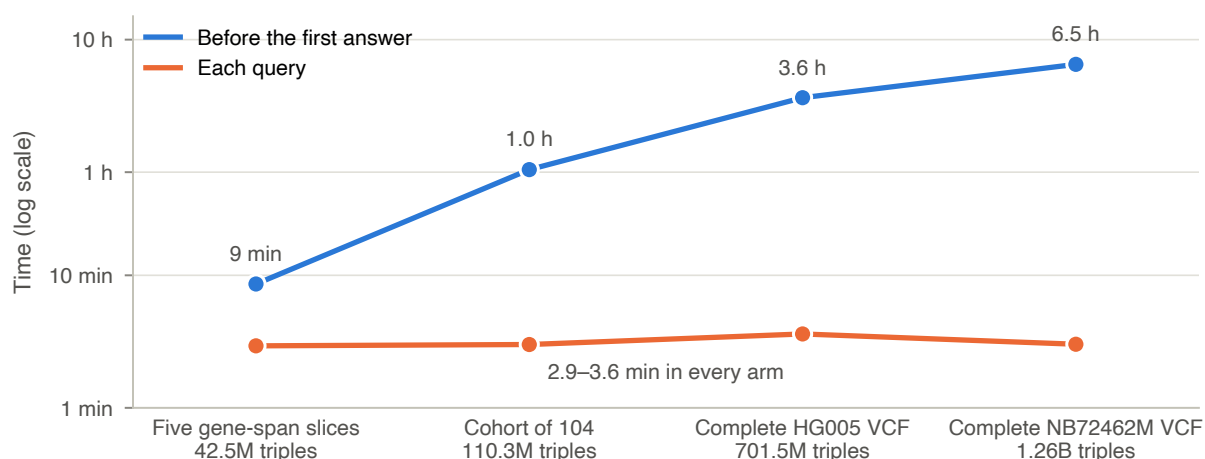


Figure S4. Most linked-workflow cost occurs before the first query. The upper line is the time before the clinical-care requester’s first answer: conversion and linking once per arm, then that requester’s view writing, validation, and indexing. The lower line is the median query time. The indexed size includes each arm’s VCF graphs, ClinVar, frequencies where applicable, and linksets.

S6 Reproducibility and execution

S6.1 The base campaign

The *base campaign* is the set of benchmark runs that produced the main text’s fidelity and scalability results: 143 runs (*cells*) in 12 experiment families, all executed with the published VCF-RDFizer v3.1.0 release on two identical hosts, **vcf-bench-1** and **vcf-bench-2** (Sections S6.2 and S6.3). Each family is one script of the benchmark harness in the testing repository, numbered as below; scripts 00 and 02 only record the environment and derive the ladder inputs, and are not counted. Every family ran as a whole on one host, never split across hosts, with no other benchmark running, and every cell wrote its command, commit, image, exit status, and timings to a fresh output directory. The families, with their inputs and what each varies, are:

Records scaling (04). Prefixes of HG005 with 10,000, 100,000, and one million records, three replicates each, and the complete VCF (3,856,856 records) once (10 cells); main Section 4.3 and Figure S5.

Sample representation (03). 10,000 records of the 1000 Genomes chromosome-20 call set at 1, 4, 16, 64, 256, 1,024, and 2,504 samples under both sample profiles (14 cells), three timing replicates at 1 and 2,504 samples in each profile (12), and four anchor cells on a single-sample and a multi-sample fixture; main Section 4.3.

Storage mode (01). Plain against space-optimized storage on 100,000 HG005 records, three replicates each, and on a 269M-triple test file once each (8 cells); main Section 4.3 and Figure S5f.

Corpus breadth (05). HDT and COTTAS construction on the first 250,000 records of eight public inputs and on the complete HG005 VCF (10 cells); main Section 4.3 and Figure S7.

Representation axes (07). Raw against structured INFO on 100,000 HG005 records and on HGSVC2, three replicates each; basic against structured header output; and fixtures declaring VCF 4.1 to 4.5 or no version (20 cells); Sections S7 and S10.

Difficult inputs (09). Eleven awkward fixtures, one conversion with the three demonstration linkers, and a custom-mapping cell (16 cells); Section S7 and Table S15.

Covering set (11). Ten configuration rows on a 1,000-line fixture that together cover every value and every pair of values of seven options, each validated with shapes (10 cells); Section S10.

Modes (12). Each operating mode run on its own on the 1,000-line fixture and its artifacts: TSV, conversion, validation, compression, decompression, and HDT and COTTAS indexing (7 cells); Section S10.

Feasibility (10). One million HG005 records under three configurations, each with requested memory ceilings of 8, 16, and 31 GB (9 cells); Section S10.

Robustness (08). Two identical conversions of 100,000 HG005 records, a convert–compress–decompress round trip, an index build and rebuild, and the 113-fault mutation score with the default and with all shape profiles (9 cells); main Section 4.1 and Section S4.

Equivalence (06). The 10,000-line (9,986-record) fixture in both storage modes and both sample profiles, validated on four SPARQL engines over three RDF artifacts, two HDT strategies, and two required refusals (8 cells); Section S10.

Query cost (13). The thirteen source-comparison queries against cyvcf2 on the 10,000-line fixture (0.96M triples) and on 100,000 HG005 records (17.1M triples), three replicates each (6 cells); main Section 4.3 and Section S9.4.

The 78 validation runs of the base campaign come from five of these families: Difficult inputs (20), Covering set (23), Equivalence (16), Query cost (18), and Modes (1).

The base campaign took 96.16 h of measured wall time (Table S12). Every asserted expectation held, but not every cell completed successfully. The truncated-gzip and sites-only fixtures exited non-zero and are recorded as outcomes; the two HDT single-strategy cells with COTTAS or with space-optimized storage were required refusals; and three cells were skipped by a guard: a 2,504-sample input under the expanded profile in two families (the 1000 Genomes call set in Corpus breadth and the multi-sample anchor in Sample representation), because its sample columns exceed the volume’s sample limit, and the custom-mapping cell, because no custom rules file was supplied. The two mutation-score cells record the host-side test-harness commit rather than a conversion-container execution.

Table S12. Base campaign by experiment family. Outcomes count cells: OK; Rec., recorded (a non-zero exit without an asserted outcome); Ref., refused (an expected rejection); Skip., skipped (a guard prevented execution). Hosts b1 and b2 are `vcf-bench-1` and `vcf-bench-2`; sections without an S prefix are in the main text.

Experiment	Host	Cells	Hours	Outcome				Question answered
				OK	Rec.	Ref.	Skip.	
<i>Conversion, scaling, and storage</i>								
Records scaling	b1	10	10.99	10				Cost growth with records (§4.3)
Sample representation	b2	30	19.69	29			1	Profile growth with samples (§4.3)
Storage mode	b1	8	12.87	8				Peak workspace, plain vs space-optimized (§4.3)
Corpus breadth	b2	10	26.90	9			1	HDT and COTTAS on ten public inputs (§4.3)
<i>Input and configuration coverage</i>								
Representation axes	b1	20	4.82	20				VCF 4.1–4.5; INFO and header costs (§S7, §S10)
Difficult inputs	b1	16	0.10	13	2		1	Eleven awkward VCFs; demonstration linkers (§S7)
Covering set	b1	10	0.35	10				Every option value and pair (§S10)
Modes	b1	7	0.03	7				Each operating mode alone (§S10)
Feasibility	b1	9	9.03	9				Memory ceiling × configuration (§S10)
<i>Validation and retrieval</i>								
Robustness	b1	9	0.25	9				Determinism, round trip, mutation score (§4.1, §S4)
Equivalence	b2	8	6.84	6		2		Four engines, both profiles, three artifacts (§S10)
Query cost	b1	6	4.27	6				SPARQL vs cyvcf2 on identical work (§4.3)
Total		143	96.16	136	2	2	3	

Runs outside the base campaign are reported separately and are not counted in its totals (Table S13): the linked workflow, regional retrieval with tabix, large-graph retrieval on `vcf-bench-3`, two additional validation runs (a default-profile mutation rerun and the consumer WGS validation run on 250,000 NG131FQA1I records, repeated with v3.3.1’s corrected validator), an N-Triples-only rerun of the 100,000-record query-cost workload, and the shared-input converter comparison. An earlier pre-release campaign and the host-calibration runs are archived with the results but not reported.

S6.2 Builds and dependencies

The base campaign used VCF-RDFizer v3.1.0, commit d3b34d5. The Docker image is `ecrum19/vcf-rdfizer:v3.1.0`, with digest

`sha256:1904e96dde12ab2e2e70d8ee1267765c293ab100a8bd1b14d5b009b2bf8e34aa`

Image rebuilding was disabled during execution. The image bundles the VCF Core SHACL shapes at vocabulary commit `5bfd19d`; this snapshot is distinct from the separately versioned vocabulary release [16].

The pinned dependencies are RMLStreamer 2.5.0, hdtc 1.1.0, pycottas 1.1.0, DuckDB 1.5.5, cyvcf2 0.34.0, pyshacl 0.30.1, Comunica 5.3.0 for N-Triples, `@comunica/query-sparql-hdt` 5.0.1, the COTTAS Comunica engine 0.1.0, bcftools from the base distribution, and QLever pinned by image digest. The image also fixes the hdt-cpp revision used to compile `rdf2hdt` and `hdt2rdf`. Later experiment families are identified separately in Table S13. Every result outside the base campaign that was first produced by a pre-release build was rerun with the published v3.3.1 release, commit `b25fb7b`, image digest

`sha256:3ad71b1a54612142e3be43b24e7e4a38551949cb641f4421bcae9bf051102993`

from one harness commit (`6f2239ff`) and the same inputs: the four use-case arms, regional and large-graph retrieval, and the consumer WGS validation run. Each ran on its original host, except Arm 4, which moved from `vcf-bench-3` to `vcf-bench-2`. Retrieval reran on the v3.1.0 graphs; converting the same 100,000 HG005 records with v3.3.1 gave sorted triples identical to v3.1.0's (17,098,746 triples). The live MyVariant.info tier did not query the service again: from harness commit `4ea07cc7`, v3.3.1 replayed the 21 responses the service returned on 2026-09-28 to the first, pre-release run, and confirmed the same links.

Table S13. Builds and assessment boundaries of each experiment family. A reproducible state requires the recorded code, local changes, dependencies, and external resources.

Family	Recorded state	Assessment boundary
143-cell base campaign	v3.1.0, <code>d3b34d5</code> ; full image digest above	Conversion, artifacts, fixtures, 984 applicable paired comparisons, and configuration tests
Regional retrieval	v3.3.1 (image <code>sha256:3ad71b1a...2993</code>) on the v3.1.0 query-cost graphs; <code>vcf-bench-1</code>	Runner with tabix; 11,160 regional executions
Large-graph retrieval	v3.3.1 on the v3.1.0 graphs; <code>vcf-bench-3</code>	Separate host; thirteen summary queries without SHACL
Release conversion check	v3.3.1 against v3.1.0, on <code>vcf-bench-2</code>	100,000 HG005 records; identical sorted triples, so the v3.1.0 graphs stand for v3.3.1's
Linked use case	v3.3.1 for every stage; Arms 1–2 on <code>vcf-bench-1</code> , Arms 3–4 on <code>vcf-bench-2</code>	SPDI, gene links, policy views, and record-level comparison; the MyVariant.info tier replays the responses recorded on 2026-09-28
Additional validation runs	v3.1.0 (<code>d3b34d5</code> , image <code>sha256:1904e96d...34aa</code>) on <code>vcf-bench-2</code>	Default-profile mutation rerun; consumer WGS validation run (250,000 NG131FQA1I records) on QLever, expanded profile, rerun without shapes under v3.1.0's size gates
Consumer WGS validation rerun	v3.3.1 on <code>vcf-bench-2</code>	Same slice and graph; QLever, expanded profile; default shapes in 117 batches on four workers
N-Triples-only rerun	v3.1.0 (<code>d3b34d5</code> , image <code>sha256:1904e96d...34aa</code>) on <code>vcf-bench-2</code>	100,000 HG005 records; conversion to gzip-framed N-Triples only, then the query-cost validation on QLever without SHACL; three replicates each
Shared-input converter comparison	Harness <code>e3743590</code> ; VCF-RDFizer v3.3.1 image <code>sha256:3ad71b1a...2993</code> ; the other converters as in Table S2, on <code>vcf-bench-1</code>	100,000 HG005 records and the 16-sample, 10,000-record ladder rung; Q1–Q8, three replicates (Section S2)

[AUTHOR QUERY: Arm 4's rows are still from its pre-release v3.3.1 run on `vcf-bench-3`; its rerun with the published image is running on `vcf-bench-2`.]

The 657.4M-triple conversion/retrieval graph and the 662.4M-triple normalized use-case graph are different artifacts, not alternative counts of the same graph.

S6.3 Hosts, measurements, and execution status

The base experiments used two separate hosts, `vcf-bench-1` and `vcf-bench-2`, each with an Intel Xeon Silver 4309Y at 2.80 GHz (8 cores), 33.6 GB RAM without swap, a 210 GB SSD, Ubuntu 24.04.4 LTS, kernel 6.8.0-85-generic, Docker 29.7.2, and Python 3.12.3. Each complete experiment ran on one host without concurrent benchmark jobs. Calibration produced identical triple sets, byte-identical COTTAS artifacts and HDT indexes, and differences only in N-Triples/HDT ordering and the HDT `dcterms:issued` timestamp. Conversion nevertheless took 26 versus 37 s, and validation 156 versus 208 s, so timings are compared within hosts. Large-graph retrieval ran separately on `vcf-bench-3`, with the same CPU, 8 cores, 31.3 GiB RAM without swap, Ubuntu 22.04.5, and kernel 5.15. It queried the v3.1.0 graphs with the v3.3.1 runner.

Each cell records its command, host, commit, image, exit status, and wall time in a fresh output directory. Stage reports record resident memory, triple counts, artifact sizes, validation outcomes, and transient workspace separately. Performance denominators use uncompressed VCF bytes; Table S14 instead reports compressed download sizes. Replicated comparisons describe variability, whereas single-run corpus results do not estimate run-to-run variation.

Commands, normalized answers, and comparison reports are maintained in the testing repository; the complete evaluated snapshots still require a persistent archive. Tagged releases are distributed through PyPI, Conda-Forge, and Docker Hub under the MIT License, with installation instructions, examples, release notes, and issue tracking [45]. The unit-test suite covers Linux, macOS, and Windows; the reported benchmarks used Ubuntu.

S7 Input and version coverage

Table S14 identifies the ten public inputs; download commands and canonical filenames are maintained in the testing repository.

Table S14. Public corpus and evaluation roles. Names are canonical `.vcf.gz` stems; assemblies and sample counts were checked against the headers. Sizes are approximate compressed downloads. WGS: whole-genome sequencing; each file is a VCF of variant calls, not a genome sequence.

File	Provider	Assembly	Samples	MB (gz)	Evaluation role
NG1N86S6FC	Sequencing.com	GRCh38	1	379	Consumer WGS
NG131FQA1I	Dante Labs	GRCh38	1	224	Consumer WGS
NB72462M	Nebula Genomics	GRCh38	1	341	Consumer WGS
60820188475559	PGP upload	GRCh37	1	325	SNP-only profile
60820188474283	Dante Labs	GRCh37	1	222	Consumer WGS
OG00R_HG002	precisionFDA	GRCh38	1	69	GIAB Truth Challenge v2
1000G_phase3_chr20	1000 Genomes	GRCh37	2,504	327	Phased cohort
HGSVC2	HGSVC2	GRCh38	32	31.5	Multi-sample SVs
HG004_GRCh38	GIAB v4.2.1	GRCh38	1	149	Benchmark VCF, chr1–22
HG005_GRCh38	GIAB v4.2.1	GRCh38	1	139	Benchmark VCF, chr1–22

Versions and corpus processing. Fixtures declaring VCFv4.1–4.5, plus one with no `##fileformat`, all converted. Their generated RML mappings differ only in the file class, from `vcfc:VCF41File` to `vcfc:VCF45File`. Version-specific fields are handled through the data: the two-record VCFv4.5 fixture carries LAA and LAD, and the VCFv4.4 fixture carries PS, yielding 252.5 and 210 triples per record, respectively. This demonstrates retention of those

fields, not interpretation of every version-specific construct; local-to-global allele resolution is not implemented. The undeclared-version input uses the documented fallback.

Nine corpus inputs were processed with headers retained: the first 250,000 records of eight files, and HG005 whole. Their HDT and COTTAS outputs passed decoding/count checks. The harness skipped the complete 2,504-sample 1000 Genomes input because its expanded-profile corpus guard excludes files with more than 1,000 sample columns. Both sample profiles were instead tested on 10,000-record slices containing all 2,504 samples. Separately, the tool refuses an expanded conversion when its estimated workspace exceeds three quarters of free space and suggests condensed output. Neither guard is evidence that the complete cohort was converted.

Difficult inputs. Nine of eleven fixtures converted and completed paired validation (Table S15). The truncated gzip was refused during decompression, while a malformed sites-only fixture produced RDF but did not complete validation. The separate sites-only frequency graph in the linking experiment demonstrates a working frequency join, not completion of the missing paired-validation test.

Table S15. Difficult-input fixtures. Triple counts describe the produced graph; an intended refusal is not a fidelity result.

Fixture	Triples	Validation	Note
CRLF line endings	262	PASS	No stray <code>\r</code> in the last field's literal
Symbolic alleles	453	PASS	<code></code> not treated as a base string
Breakend ALT	417	PASS	Breakend ALT syntax preserved
Mixed ploidy	350	PASS	Ploidy 4 then 3 followed the data
Half and haploid calls	446	PASS	Four genotype cases stayed distinct
Missing values	321	PASS	Missingness follows the field-specific representation
FILTER variants	486	PASS	PASS, ., and a failed code are three states
Declared FORMAT unused	293	PASS	Header declaration represented although never used
Zero data records	189	PASS	Header-only input
Sites only	245	incomplete	Graph and HDT artifact produced; validation did not complete
Truncated gzip	—	refused (intended)	Refused at decompression in 0.58 s with a diagnostic

S8 Defects, corrections, and remaining gaps

Table S16 records defects and incomplete checks that are not visible in aggregate pass counts. Corrected outputs are used where stated, while unsuccessful runs remain in the evidence record.

Table S16. Defects, diagnostic evidence, and current status.

Issue	Evidence	Correction or remaining limit
Header-only samples	Q9 and Q10 showed that a file with declared samples but no records omitted its <code>SampleSet</code> and <code>VCFSample</code> resources.	Sample resources are now derived from the header, increasing the fixture from 181 to 189 triples; the preflight oracle was also corrected to expect no per-record calls.
Dangling allele references	Raw INFO with expanded samples produced 1,972 genotype references to absent allele resources and 5,916 shape violations in a 1,000-record test.	Alleles are emitted whenever INFO or sample representation requires them; census expectations and regression tests cover all four INFO/sample combinations.

Table S16, continued

Issue	Evidence	Correction or remaining limit
Withheld-file metadata	The policy plug-in removed the correct records but retained a withdrawn file’s header.	Ownership rules now define the dependent context removed with a withheld resource.
Shape memory guards	A 170,935,101-triple graph was admitted as 390.7 MB COTTAS and exhausted 30.7 GiB, while larger packaged forms of the same graph were skipped; a 58.2M-triple graph was also admitted by its 63.8 MB VCF size.	Version 3.3.0 gates by triples, and v3.3.1 validates record-local default shapes in batches; deeper cross-record profiles retain a whole-graph limit.
Comunica heap limits	Comunica-backed endpoints exhausted the JavaScript heap while approximately 25 GB of host memory remained.	The heap ceiling is configurable; the failed runs remain archived.
COTTAS query execution	The original pycottas/RDFLib path stalled on Q5 for approximately 41 and 10 hours, and in-process timeouts did not interrupt it.	The DuckDB-backed Comunica path reduced Q5 to 2–5 seconds and the 13-query workload to 38 seconds; completed comparisons use this path.
SHACL severity and missing tokens	The report parser treated recommendations as blocking, and the missing-token rule flagged legitimate dots in variable-cardinality declarations and missing genotypes.	Only <code>sh:Violation</code> blocks a run, and legitimate dot uses are excluded; nine non-malformed flags remain in the v3.1.0 campaign record.
Sites-only fixture	The fixture declared FORMAT without samples, which bcftools rejects; conversion produced 245 triples and HDT, but paired validation did not complete.	No pass or semantic failure is claimed; a conformant fixture remains necessary.
Legacy measurements	An earlier benchmark under-reported partitioned compression time, labeled uncompressed input bytes as <code>.vcf.gz</code> , and compared HDT with uncompressed rather than gzip-framed N-Triples.	Timing was corrected in commit <code>fb48e50</code> ; the legacy tables are excluded from the current analysis.
Consumer WGS oracle	Version 3.1.0 omitted phase sets and compared lexical QUAL forms, producing three validator discrepancies although direct record fields agreed.	Version 3.3.1 corrected both and added expectations for structural-variant events, confidence intervals, and gVCF blocks; local-allele semantics remain outside the oracle.

Table S16, continued

Issue	Evidence	Correction or remaining limit
Gene links for non-explicit alleles	Version 3.3.0 skipped spanning-deletion records, preventing gene-based policies from selecting them; symbolic alleles and breakends also lack links, so v3.3.1's Arm 2 views released 110 (DS) and 58 (GRU) symbolic structural-variant records in restricted genes.	Version 3.3.1 links spanning deletions by REF span; symbolic and breakend extents remain an explicit limitation.

S9 Conversion, storage, and retrieval measurements

S9.1 Record and sample scaling

On the HG005 record ladder, triple production had a fitted exponent of $+0.999$ (95% CI $[+0.991, +1.007]$), with 170.9–171.9 triples per record (Figure S5). Artifact bytes, peak workspace, and end-to-end time had fitted exponents of $+1.033$, $+1.025$, and $+1.063$. End-to-end times were 40.7, 423.7, and 5,476.0 seconds at 10,000, 100,000, and one million records, with replicate standard deviation at most 13.5 seconds. Mapping-stage resident memory remained 1.0–1.7 GB. HDT construction increased from 38% of total time at 10,000 records to 82% for the complete VCF.

The sample ladder holds 10,000 records and INFO fixed while varying samples from one to 2,504 (Figure S6). Triple counts are exactly $1,359,498 + 250,005S$ expanded and $1,439,498 + 5S$ condensed. At 2,504 samples, expanded output contains 627,372,018 triples, or $432\times$ the condensed graph. The byte ratios are $322\times$ for gzip-framed N-Triples and $69\times$ for HDT. Condensed HDT grows $5.8\times$ as vectors lengthen despite only 0.9% triple growth. Expanded HDT reaches 4.10 GB and requires 15,269 seconds, compared with 59.2 MB and 250 seconds for condensed output. The projected expanded workspace for the complete 1,812,841-record call set is approximately 2 TB, so the run was refused; the projection is not a completed full-cohort measurement.

S9.2 Storage mode and optional artifacts

Plain and space-optimized storage produced identical triples and final artifact sizes within one filesystem block. Space-optimized mode reduced peak workspace from 2.98 to 0.325 GB on 100,000 records and from 43.40 to 5.65 GB on a 1,155,741-record prefix, with time increases of 5.2% and 3.2%. Mapping time met a prespecified $\pm 10\%$ equivalence margin on the replicated input.

Across the corpus, COTTAS occupied 0.37–0.55 times gzip-framed N-Triples and 0.27–0.37 times HDT, while taking 1.5–3.0 times longer than HDT to build (Figure S7). The v3.1.0 complete-HG005 run produced 657.4 million triples in 16.07 hours, including approximately 14.5 hours for HDT and COTTAS construction. The retained HDT, HDT index, and COTTAS tree occupied 10.88 GB, while gzip-framed N-Triples alone occupied 2.89 GB. The later v3.3.1 use-case conversion produced 662.4 million genomic triples in 53 minutes, or 55 minutes with ClinVar, before linking, release processing, and indexing. The measurements differ in version, input, and work performed and are not estimates of a speed-up.

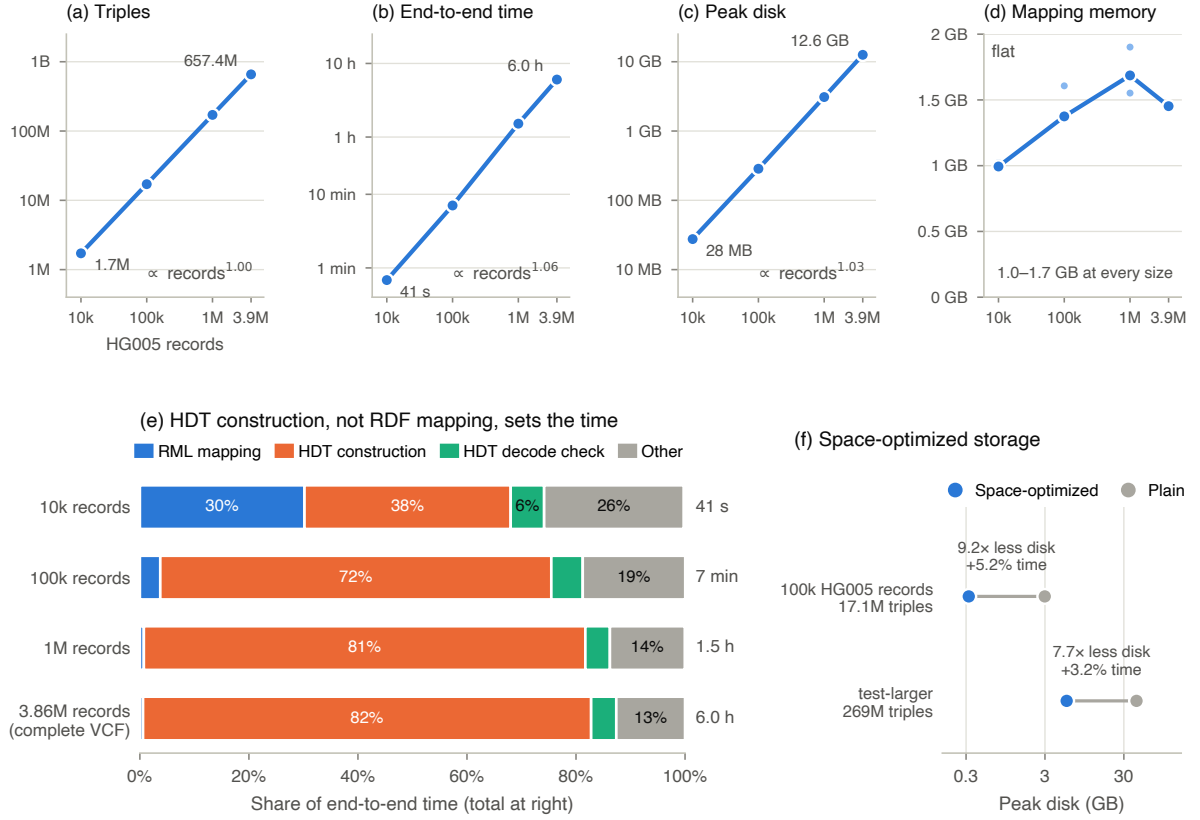


Figure S5. Record scaling and storage mode. Panels (a)–(e) report triples, time, peak disk, mapping memory, and stage shares for the HG005 record ladder. Panel (f) compares plain and space-optimized workspace. The three ladder rungs are replicated, while the complete VCF is one run.

S9.3 Query workloads and physical access paths

On 17.1 million triples, QLever’s thirteen-query totals from N-Triples, HDT, and COTTAS differed by less than 1%, because each path queried a QLever index rather than the artifact natively (Figure S8). On a separate 0.96-million-triple graph, mean totals were 1.1 seconds for QLever, 23.4 seconds for Comunica over N-Triples, 38.2 seconds for COTTAS, and 45.1 seconds for HDT, with identical answers.

For individual questions on 17.1 million triples, full cyvcf2 parses required 11.5–12.3 seconds, while indexed QLever required 0.005–4.62 seconds. The query-time ratios exclude conversion and the 22.8-second index build. One cyvcf2 pass answered all thirteen questions in 12.4 seconds, compared with 17.1 seconds of indexed queries before setup.

The complete-HG005 workload required 646.4 seconds for all thirteen questions after an 880.8-second index build, with every answer equal to the parser result. The three QLever input artifacts differed by only 0.4% in indexed query time but differed in decoding time. Native HDT and COTTAS paths answered five questions and failed on Q6 at 171 million triples despite a 24 GB heap. Small-graph engine rankings therefore did not predict large-graph completion.

Indexed regional access. Across 11,160 regional executions, every answer agreed and no execution failed. On 17.1 million triples, QLever required 10.3–10.8 milliseconds across 1 kb–10 Mb windows. Tabix-indexed cyvcf2 and bcftools were faster at 1 kb but grew with the number of records parsed; QLever led both at 10 Mb. QLever setup required 23.3 seconds, compared with 0.46 seconds for bgzip and tabix. Figure S9 separates the five regional questions and reports per-question break-even counts.

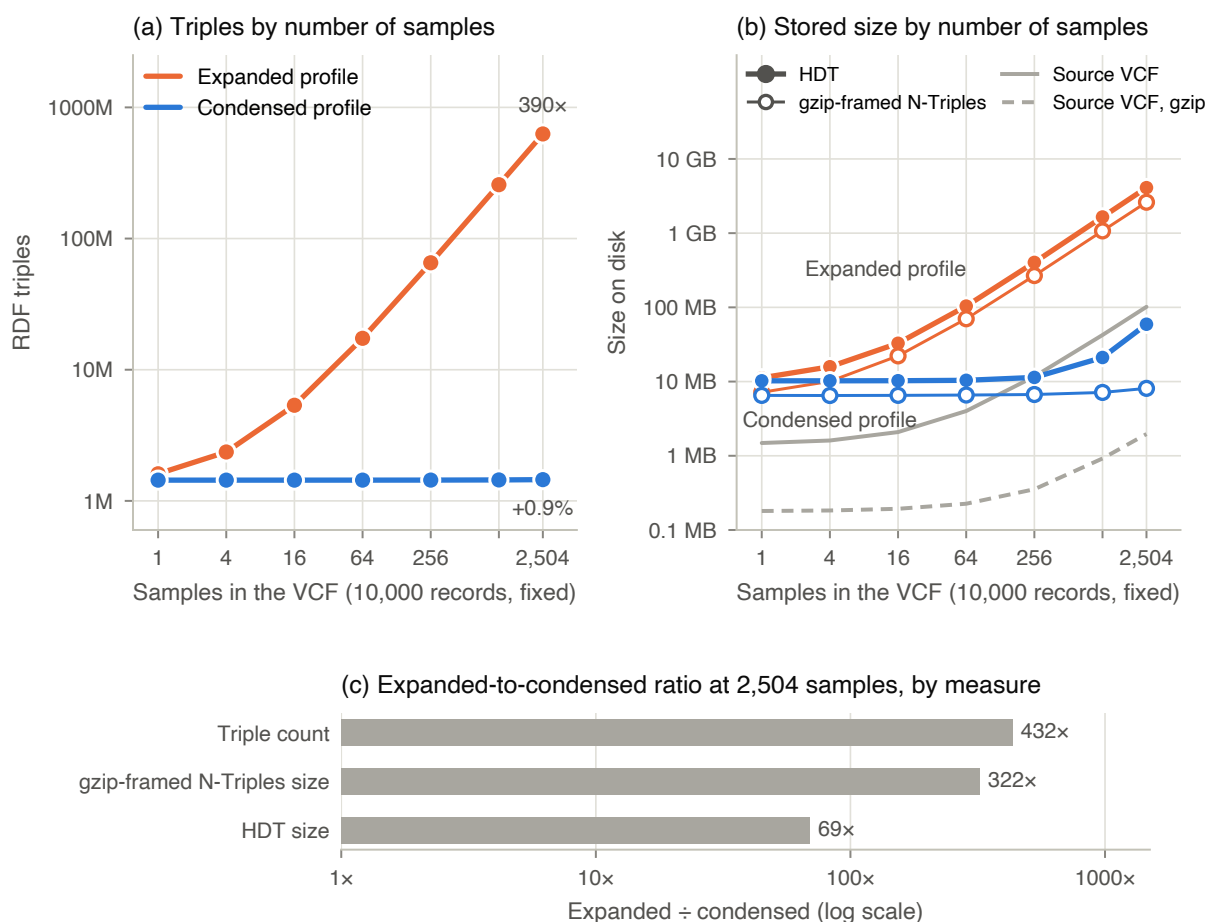


Figure S6. Sample representation trades direct addressability for size. The same 10,000 records are converted with one to 2,504 samples. Expanded resources grow with sample calls, while condensed vectors keep triple counts nearly constant but still grow in bytes.

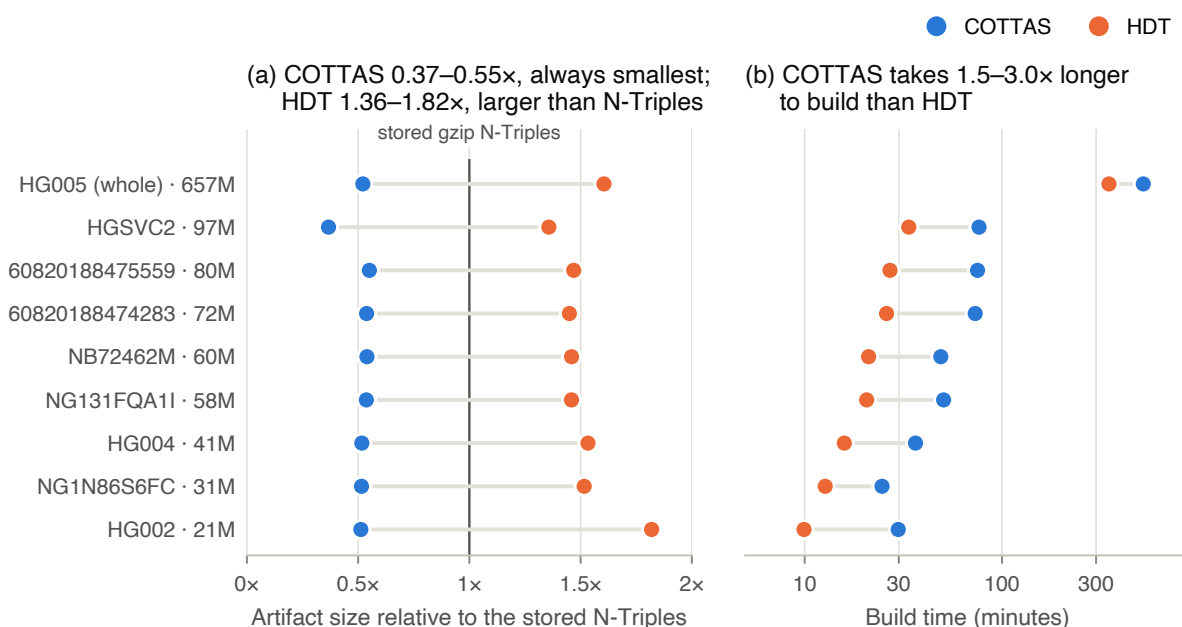


Figure S7. Corpus artifact sizes and construction times. Eight inputs use 250,000 records, and HG005 is processed completely. Artifact sizes are relative to gzip-framed N-Triples, and construction times use a logarithmic scale.

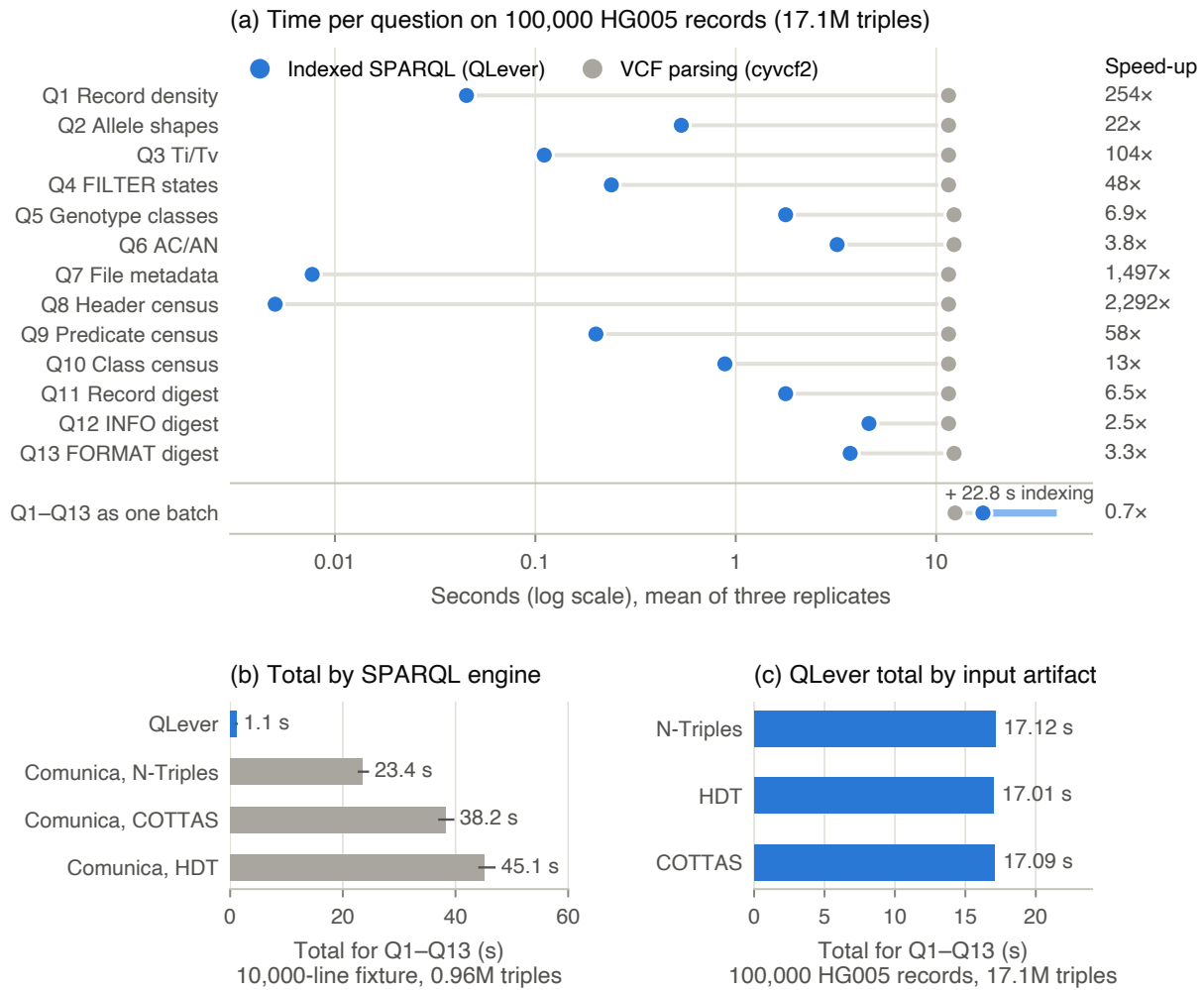


Figure S8. Detailed retrieval paths. Panel (a) compares individual questions and the batch on 17.1 million triples. Panel (b) compares four engines on 0.96 million triples. Panel (c) loads the same 17.1-million-triple graph into QLever from N-Triples, HDT, and COTTAS.

Table S17. Retrieval at scale on v3.1.0 HG005 graphs. Smaller graphs ran on `vcf-bench-1` with v3.1.0 and larger ones on `vcf-bench-3` with v3.3.1, so normalized times keep host differences. Completed answers were compared with `cyvcf2`; setup is reported separately.

Graph	Triples	Engine	Time (s)	Outcome
<i>Does SPARQL scale to a complete VCF? (all thirteen questions)</i>				
Fixture	958,919	QLever	1.1	1.13 s per million triples
100 k records	17,098,746	QLever	17.0	1.00 s per million
1 M records	170,935,101	QLever	165.1	0.97 s per million
3.86 M records, complete	657,425,805	QLever	646.4	0.98 s per million; setup 880.8 s; replicates within 0.6%
<i>Does the artifact matter? (complete VCF, one engine, three packagings)</i>				
from N-Triples	gzip-framed		646.4	Decode 184–187 s; setup 880.8 s
from HDT	—		644.8	Decode 963–979 s; setup 887.8 s
from COTTAS	—		643.9	Decode 600–608 s; setup 891.3 s
<i>Do the other engines get there at all? (1 M records, 170,935,101 triples)</i>				
COTTAS engine	5 of 13 answered		751.4	Native allocation failure on Q6
HDT engine	5 of 13 answered		12,489.5	Q5 answered at the one-hour per-query ceiling (3,600.1 s); Q6 failed

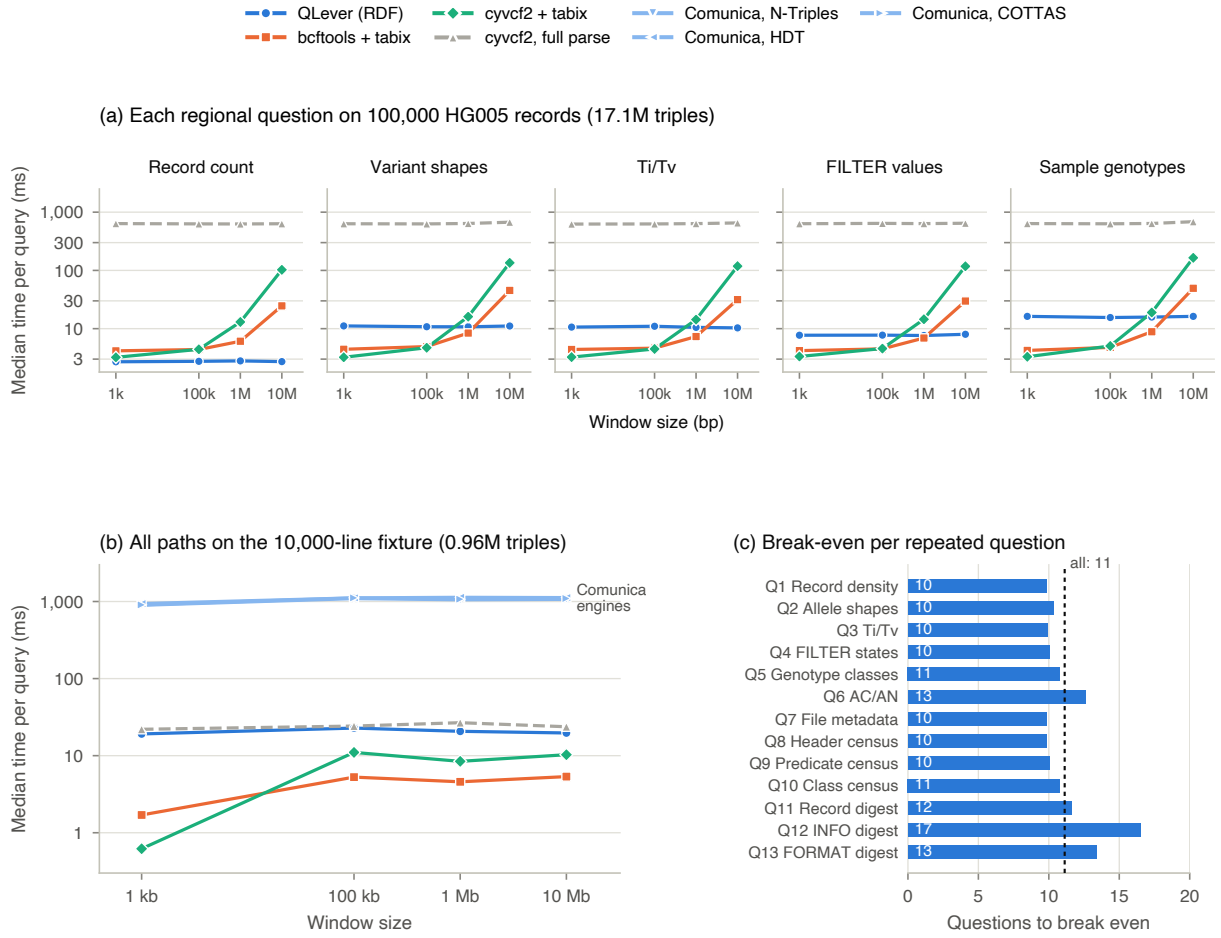


Figure S9. Regional and repeated-question retrieval in detail. Panel (a) shows each regional question on 100,000 HG005 records. Panel (b) compares seven execution paths on the 10,000-line fixture. Panel (c) reports the break-even repetition count for each whole-file question, including minimal RDF setup.

Table S18. Regional retrieval on the 17.1M-triple graph of 100,000 HG005 records, excluding setup; the data behind main Figure 5c. Entries are medians in milliseconds over five questions' window medians and three repetitions. Tabix-indexed region queries use twenty windows per size; full parses of the unindexed VCF use three. All paths use the same POS-based selection; the bottom row gives the median records per window.

Execution path	1 kb	100 kb	1 Mb	10 Mb
QLever (RDF)	10.6	10.8	10.5	10.3
bcftools + tabix	4.2	4.6	7.3	31.5
cyvcf2 + tabix	3.2	4.6	14.5	118.2
cyvcf2, full parse	633.3	629.2	635.8	653.4
<i>Records per window</i>	<i>2</i>	<i>143</i>	<i>1,262</i>	<i>13,143</i>

S9.4 Setup amortization

For an equivalent repeated workload with a positive per-question saving,

$$N_{\text{break-even}} = \frac{\text{additional RDF setup}}{\text{VCF time per question} - \text{indexed RDF time per question}}. \quad (2)$$

A three-replicate rerun measured the minimal QLever path on 100,000 records. Conversion to gzip-framed N-Triples required a mean 95.7 seconds, QLever indexing 25.0 seconds, a full cyvcf2 parse a median 12.28 seconds per question, and an indexed query a mean 1.43 seconds. The resulting crossover is approximately eleven separately executed questions. Including the unnecessary HDT build raises setup to 446.5 seconds and the corresponding estimate to 44 questions. The calculation assumes a full VCF parse for each question at the measured query mix, rather than one batch pass. Per-question crossovers range from 10 to 17 repetitions. The gzip-framed N-Triples occupied 75.8 MB versus a 3.1 MB gzipped VCF for 100,000 records and 2.89 GB versus 139.3 MB for complete HG005. No universal break-even count follows, and update costs for changing evidence or policies were not measured.

S10 Configuration coverage, operating modes, and memory feasibility

A ten-row covering set exercised every value of seven axes: sample representation, INFO representation, storage mode, RDF compression, output representations, artifact compression, and HDT strategy. Every row completed validation, including shapes. Pairwise coverage excludes only three prohibited combinations: `--hdt-strategy single` with space-optimized storage, COTTAS, or `hdt,cottas`. Gzip and Brotli were exercised in both compression roles, but their ratios and throughput were not compared.

The `tsv`, `compress`, `decompress`, HDT-index, COTTAS-index, conversion-only, and validation-only modes each ran independently with exit code 0. A feasibility sweep crossed requested 8, 16, and 31 GB memory ceilings with three configurations on one million records. All nine cells completed at 1.34–1.93 GB resident memory, but the harness could not confirm that Docker limits were forwarded to all container invocations. The result therefore shows low measured memory for those runs, not survival under verified enforced limits.

Structured INFO increased the structural-variant graph from 89.01M to 96.75M triples (8.7%), N-Triples bytes by 7.4%, and HDT bytes by 7.8%. On 100,000 HG005 records, the increase was from 12.52M to 17.10M triples (36.6%). Raw INFO still retains alleles needed by expanded sample calls after the dependency correction in Section S8. Structured header output added only 0.03%: 17,098,746 versus 17,094,293 triples.

The cross-encoding experiment used a 10,000-line, 9,986-record single-sample fixture in both sample profiles and storage modes, with N-Triples, HDT, and COTTAS artifacts. All sixteen validations passed the fourteen preflight checks and Q1–Q13 under all four engines, on graphs of 0.96M triples expanded and 0.79M condensed. The complete condensed cell took 70.4 min. Two additional negative tests correctly refused single-HDT strategy with COTTAS or space-optimized storage, returning exit status 2.

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